Climate inequality

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Abstract

This paper characterizes the relationship between anthropogenic climate change and economic inequality within countries, a subject which intersects two of the defining social challenges of the 21st century but remains remarkably understudied. To address this urgent research gap, I build on cutting-edge methods for estimating climate impacts and apply these refinements to newly available distributional national accounts data. I document new evidence that temperature shocks exacerbate income inequality within countries, an effect driven by disproportionate reductions in income among the poorest segments of already-poor countries. To better understand the implications of this incidence pattern for the welfare economics of climate change, I then integrate these results over observed income distributions and use output from the newest generation of global climate models to perform a counterfactual analysis estimating anthropogenic contributions to these climate inequalities. These findings represent the most comprehensive characterization yet of the economic regressivity of climate change within countries and globally.

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